

KUMBAM PARIKSHITH REDDY

Software Engineer | Backend • Cloud • AI/ML

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PROFILE

Backend-focused Software Engineer with 1+ year of full-time professional experience, working primarily with Java, Spring Boot, SQL/databases and production systems, with growing expertise in cloud and distributed systems, alongside AI/ML. Through professional work and personal projects, I have developed a strong interest in building scalable backend systems and integrating intelligent/ML-driven capabilities into them. I now want to pursue a Master's in Germany to deepen my knowledge of AI/ML while continuing to build expertise in backend and cloud engineering.

EDUCATION

Neil Gogte Institute of Technology, Osmania University — Hyderabad, India

2021-2025

Bachelor of Technology in Computer Science and Engineering
CGPA: 7.56

PROFESSIONAL EXPERIENCE

SplashBI — Associate Software Engineer | Hyderabad, India | 1+ year

AUG 2025-PRESENT

- Independently investigate and resolve 20+ production issues per month, including application failures, report execution problems, synchronization issues, and incorrect-data incidents across the application.
- Diagnose backend and production issues using Java, Spring Boot, MySQL/SQL, application logs, HAR files, and Tomcat logs, identify root causes, and implement corresponding code fixes.
- Develop fixes for business-logic defects, exception/error-handling issues, performance problems, and approved product enhancements, validating changes locally before committing them to Git and handing them to QA.
- Optimized a slow-running report by investigating application logic and database behavior, reducing execution time from approximately 2 hours to 1 hour.
- Investigated scheduled-report date/time discrepancies across database and server-level timezone configurations, identified the root cause, and provided a corrective solution.
- Perform database-level troubleshooting including SQL query development, incorrect-data investigation, data correction, table and index changes, joins, subqueries, stored procedures, query optimization, and database error analysis.
- Collaborate with 3-5 engineers, QA, developers, and product/business teams to assess enhancement feasibility, review fixes, validate defects, and coordinate resolutions.
- Resolved an urgent client-facing application outage by analyzing logs during a live support call, deploying the corrected class file to the affected instance, and clearing application cache to restore application functionality.

SELECTED PROJECTS

Workflow Orchestration Engine | Java 21 | Spring Boot | PostgreSQL | Redis | Kafka | Quartz | Docker | [GitHub](#)

- Designed and implemented a production-style workflow orchestration engine supporting versioned workflow definitions, lifecycle management, and execution through explicit workflow and task state machines.
- Implemented 10 typed task handlers covering decision, script, parallel, notification, service, HTTP, webhook, approval, delay, and human tasks, with invalid state transitions explicitly rejected.
- Integrated Apache Kafka for workflow lifecycle event streaming and audit logging, Quartz for cron-based scheduling and delayed execution, and Redis for distributed locking, rate limiting, and workflow-definition caching.
- Built a JWT-secured REST API with role-based authorization and centralized exception handling, supported by JUnit, Mockito, AssertJ, and Testcontainers integration testing.
- Achieved 83%+ line coverage with JaCoCo and configured GitHub Actions CI to execute the test suite and enforce coverage gates on every push and pull request.
- Containerized the complete application stack using Docker and Docker Compose with PostgreSQL, Redis, Kafka, and the Spring Boot application.

Campus Workforce Management Platform | Java 21 | Spring Boot | Spring Cloud Gateway | MySQL | Spring Security | JWT | Docker | [GitHub](#)

- Architected and implemented a 16-service microservices platform covering campus employment workflows including job posting, applications, scheduling, attendance, payroll, leave, performance, certificates, notifications, training, documents, and analytics.
- Designed each service with an isolated MySQL database and authenticated REST-based communication, avoiding shared database access across service boundaries.
- Implemented Spring Cloud Gateway as the unified API entry point and developed JWT-based authentication, role-based authorization, and ownership-based access control across services.
- Built cross-service workflows including attendance-based payroll calculation and certificate generation while preserving historical rate and hour snapshots for data integrity.
- Containerized all backend services using Docker Compose and validated end-to-end service interactions, authorization rules, cross-service validation, and duplicate-prevention constraints.

Personality-Aware Product Recommendation Platform | Java 21 | Spring Boot | Python | FastAPI | XGBoost | SHAP | LIME | PostgreSQL | Redis | [GitHub](#)

- Developed a full-stack personality-aware recommendation platform combining a Java/Spring Boot backend with a Python/FastAPI machine-learning microservice to generate personalized recommendations across six domains: products, books, courses, careers, movies, and music.
- Implemented a Big Five personality assessment pipeline using the IPIP-50 questionnaire, feature engineering, cross-validation, hyperparameter tuning, and XGBoost-based prediction with MLflow experiment tracking.
- Designed a hybrid recommendation engine combining rule-based scoring with ML confidence and implemented SHAP/LIME explainability to generate human-readable reasoning for ML-backed recommendations.
- Implemented JWT authentication, role-based access control, assessment history, recommendation feedback, saved items, administrative analytics, and model-management functionality.
- Built automated unit and integration testing using JUnit, Mockito, Spring MockMvc, PyTest, and PostgreSQL, and containerized the backend, ML service, database, and Redis using Docker Compose.

AI Resource Scheduler | Java 21 | Spring Boot | Python | FastAPI | scikit-learn | PostgreSQL | SHAP | Docker | [GitHub](#)

- Built an AI-powered scheduling platform that ranks employees for open shifts using a trained gradient-boosting model based on skill match, availability, workload, and historical performance signals.
- Developed a Spring Boot backend and Python/FastAPI ML service architecture, with the backend managing employees, shifts, assignments, authentication, and recommendation history.
- Implemented SHAP-based explanations for recommendations and incorporated workload history and availability constraints to improve scheduling fairness and prevent conflicting assignments.
- Added JWT-secured REST APIs, Swagger/OpenAPI documentation, JUnit/Mockito and PyTest testing, Docker Compose containerization, and GitHub Actions CI.

TECHNICAL SKILLS

Programming: Java, Python, SQL, JavaScript

Backend: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, REST APIs, Microservices, Spring Security, JWT

Distributed Systems: Apache Kafka, Redis, Quartz, Inter-service REST Communication

Databases: MySQL, PostgreSQL

Cloud & Infrastructure: AWS, Docker, Docker Compose, GitHub Actions

AI/ML: Machine Learning, scikit-learn, XGBoost, FastAPI, SHAP, LIME, MLflow, Explainable AI

Testing & Development: JUnit 5, Mockito, AssertJ, Testcontainers, PyTest, Maven, Git

API & Tools: Swagger, OpenAPI, Linux